PACKT EVENT SERIES • SESSION 2 OF 4

AI-Powered Excel Workflows

Cut the manual spreadsheet work — clean, build, and check finance data faster with Claude.

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Early July 2026 • Live cohort on Airmeet • 2 hours

SESSION 1 RECAP

The AIPA Foundations

The HACK Framework

Hygiene → Automation → Capability → Knowledge. Sequential, not parallel.

Six Pillars

Excel Fluency, Process & Data, Tech Assessment, AI Concepts, Growth Mindset, Human Skills.

Systems Thinking

Think end-to-end, not task-level. Understand structure before changing numbers.

Probabilistic vs Deterministic

AI is not a calculator. Generate → Verify → Authorise. No AI on money-moving steps.

How Session 2 maps to the frameworks

HACK Stages in Play

- H — Clean messy data & fix formats
- A — Formula writing & batch automation
- C — Build & verify models with AI
- K — Interpret outputs & apply judgement

Pillars We Exercise Today

- **1. Excel Fluency** — formulas, data cleaning
- **2. Process & Data** — ETL, reconciliation
- **4. AI Concepts** — prompting for Excel tasks
- **5. Growth Mindset** — new ways of working

WHERE WE ARE

Session 2 of 4 — building on Foundations

From Session 1

- Claude set up safely for finance
- Model training switched off
- You know what data is in-bounds
- Finance-safe prompting framework

Today — Excel

- Get finance data into Claude cleanly
- Clean messy spreadsheets fast
- Write & audit formulas in plain English
- Build mini-models and reconciliations
- Automate summaries and commentary

BY THE END OF TODAY

What you'll be able to do

Clean messy data

Fix mixed formats, text-as-numbers, duplicates and stray characters in seconds.

Write formulas in plain English

Describe the result you want; get a working SUMIFS, XLOOKUP or variance flag.

Build & audit models

Stand up a small model from a prompt, and explain or debug existing ones.

Automate reporting

Turn raw rows into summaries and variance commentary you can reuse monthly.

PART 1

Getting Data In

Move finance data into Claude the right way — clean inputs make every later step easier.

Your data to Claude — or Claude to your data

Upload a file

- Attach .xlsx, .csv or .pdf
- Best for full tables & datasets
- Claude reads structure and headers
- Keep it de-identified

Paste a range

- Copy cells straight from Excel
- Great for quick, small checks
- Fast back-and-forth on a formula
- Watch row/column alignment

Share a screenshot

- For layouts or error messages
- Claude reads numbers from images
- Handy on locked-down machines
- Double-check transcribed figures

THE BETTER PATH | Bring Claude to your data — the M365 add-in runs natively inside Excel. No uploading, no copy-paste. A true AI agent working where you already work.

Common finance messes — and the fix

Numbers stored as text	*"Convert this column to real numbers and flag any that won't parse."*
Inconsistent dates	*"Standardise every date to DD/MM/YYYY and list the rows you changed."*
Duplicates & blanks	*"Find duplicate invoice rows and empty cells, then summarise what you found."*
Merged / messy headers	*"Reshape this into a clean table with one header row I can pivot."*

Always ask Claude to tell you what it changed — keep a human eye on every transformation.

PART 2

Formulas & Functions

Stop wrestling with syntax — describe the outcome and let Claude write and explain the formula.

FORMULAS

Plain English in, working formula out

"Sum sales where region is APAC and month is June."	`=SUMIFS(Sales, Region, "APAC", Month, "Jun")`
"Look up the budget for this account code."	`=XLOOKUP(Code, Codes, Budget)`
"Flag any variance worse than 10% vs budget."	`=IF(ABS(Var)>0.1, "Review", "OK")`
"Show a rolling 3-month average of revenue."	`=AVERAGE(OFFSET(...,-2,0,3,1))`

Describe the business outcome, not the syntax — then test the formula on a couple of known rows.

FORMULAS

Explain & audit what you inherited

Decode a monster formula

Paste a nested IF/INDEX/MATCH and ask Claude to explain it step by step.

Find the error

"Why is this returning #REF!? Show me the broken reference and the fix."

Sense-check the logic

Ask whether the formula actually does what the label claims it does.

Document it

Get a plain-English note you can drop beside the cell for the next person.

PART 3

Building & Modelling

From a blank sheet to a structured, checkable model — with reconciliations built in.

Stand up a model from a prompt

Ask for structure

"Build a 12-month revenue model with assumptions for growth and churn in separate input cells, and totals that flow through."

- Assumptions live in their own cells
- Scenarios you can flex
- Totals as formulas, not hardcodes
- Clear labels and units

Excel still owns the maths — Claude drafts the structure, you keep the calculation engine.

BUILDING

Reconciliations & checks — the Checker pattern

Tie out two lists	"Compare these two ledgers and list anything that doesn't reconcile, with the row."
Hunt for anomalies	"Flag rows where the amount looks unusual versus the rest of the column."
Re-perform the total	"Recompute this subtotal independently and tell me if it matches."
Build a check column	"Add a column that returns OK or Review against these three rules."

PART 4

Automating Reporting

Turn the same monthly grind into a repeatable, minutes-not-hours workflow.

AUTOMATING

From raw rows to a finance summary

Summarise raw data

"Group this by region and month and give me totals and % of revenue."

Draft variance commentary

"Write three bullets explaining the biggest movements vs budget."

Build the board lines

"Turn this into five plain-English lines for a board pack."

Hand it back to Excel

Drop the cleaned table or formulas straight back into your workbook.

Make it repeatable for the team as skills

Save skills	Keep the prompts that worked as skills — a Cleaner, a Checker, a Commentary writer.
Share a team library	Agree a common set so everyone gets consistent, reviewable output.
Reuse month on month	Same prompt, new file — the monthly close gets faster every cycle.
Keep the guardrails	De-identified data in, human sign-off out — every single time.

Live demo & common pitfalls

We'll do it live

- Upload a messy sample workbook
- Clean and standardise the data
- Write a SUMIFS + variance flag
- Generate variance commentary
- Drop it all back into Excel

Watch out for

- Trusting totals without re-checking
- Pasting confidential or named data
- Letting Claude hardcode the maths
- Vague prompts, vague spreadsheets
- Skipping the human sign-off

Building with AI: 4 Phases, 4 Skills

From starting file (6 sheets) to complete solution (17 sheets)

Phase 1
Data Validation
57 automated checks

Phase 2
Management Reports
SUMIFS vs LAMBDAs

Phase 3
Forecast Accuracy
H1 CY2026 comparison

Phase 4
Scenario Analysis
5 interconnected sheets

/training-data-reconciliation

Phase 1: Data Validation & Reconciliation

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- 57 PASS/FAIL checks across 8 sections
- Formula patterns: ROWS, SUMPRODUCT, COUNTIFS, SUMIFS
- Validates: row counts, roll-forward, table conversion, report tie-out
- Conditional formatting: Green PASS / Red FAIL
- Design principle: Expected (hardcoded) vs Actual (formula) = PASS/FAIL

Phase 2: Management Reports — Two Approaches

- Table Report: SUMIFS — explicit, one formula per cell (432 formulas)
- Dynamic Report: PLDATA/PLDATABY — one formula spills entire block
- Both produce IDENTICAL output
- Teaching contrast: Verbose-but-clear vs Concise-but-abstract
- 4 entities × 3 views × 12 P&L lines × 36 months

Phase 3: Forecast Accuracy — Are We On Track?

- H1 CY2026: 6-month overlap period (Jan–Jun 2026)
- FILTER isolates the date range dynamically
- Actuals vs Budget vs Forecast comparison
- Accuracy % = 1 − ABS(Variance / Benchmark)
- Green ≥95%, Red <80%

Phase 4: Scenario Analysis & Executive Summary

- Scenario Manager: 11 drivers × 3 scenarios (Base/Upside/Downside)
- Driver Analysis: Volume × Rate decomposition, 18-month window
- Forecast Output: Monthly P&L linked to drivers
- Scenario Dashboard: Toggle switches active scenario instantly
- Executive Summary: KPIs, FY P&L, entity breakdown, 4 charts
- All forecasts reference Scenario Manager — one change updates everything

Workbook Architecture: 17 Sheets, Zero Hardcodes

Source Data
6 Starting Sheets

Dynamic Extracts
LAMBDAs

Reports

Accuracy

Scenarios

Executive Summary

Validation Checks validate every connection

100% formula-driven outputs

Workflows worth keeping from this session

Each one saves 30–90 minutes per cycle when prompted well

Formula Generation

- Describe logic in plain English → working formula
- SUMIFS, INDEX/MATCH, dynamic arrays
- Saves ~20 min per complex formula

Data Cleaning & Transformation

- Messy CSV/PDF → structured Excel table
- Consistent formatting, deduplication, validation
- Saves ~60 min of manual cleanup

Workbook Audit & Debugging

- Find hardcodes, broken refs, circular logic
- Validation checks that flag errors automatically
- Saves ~45 min vs manual trace-through

Financial Model Scaffolding

- Assumptions → drivers → 3-way model skeleton
- COA mapping and forecast logic structure
- Saves ~90 min on model setup alone

RECAP

Your Excel workflow checklist

- Data brought in clean & de-identified
- Messy formats fixed and logged
- Formulas written in plain English
- Inherited formulas explained & audited
- Model structure with separate assumptions
- Reconciliation / Checker column in place
- Variance commentary drafted
- A human reviews every number

THANK YOU

Spreadsheets, minus the grind.

Next: Session 3 — Word + PowerPoint Reporting

Mid July. Turn your numbers into board-ready narrative and slides.

Before next time

Clean one real (de-identified) workbook with a saved prompt.

Questions? Let's open it up.

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